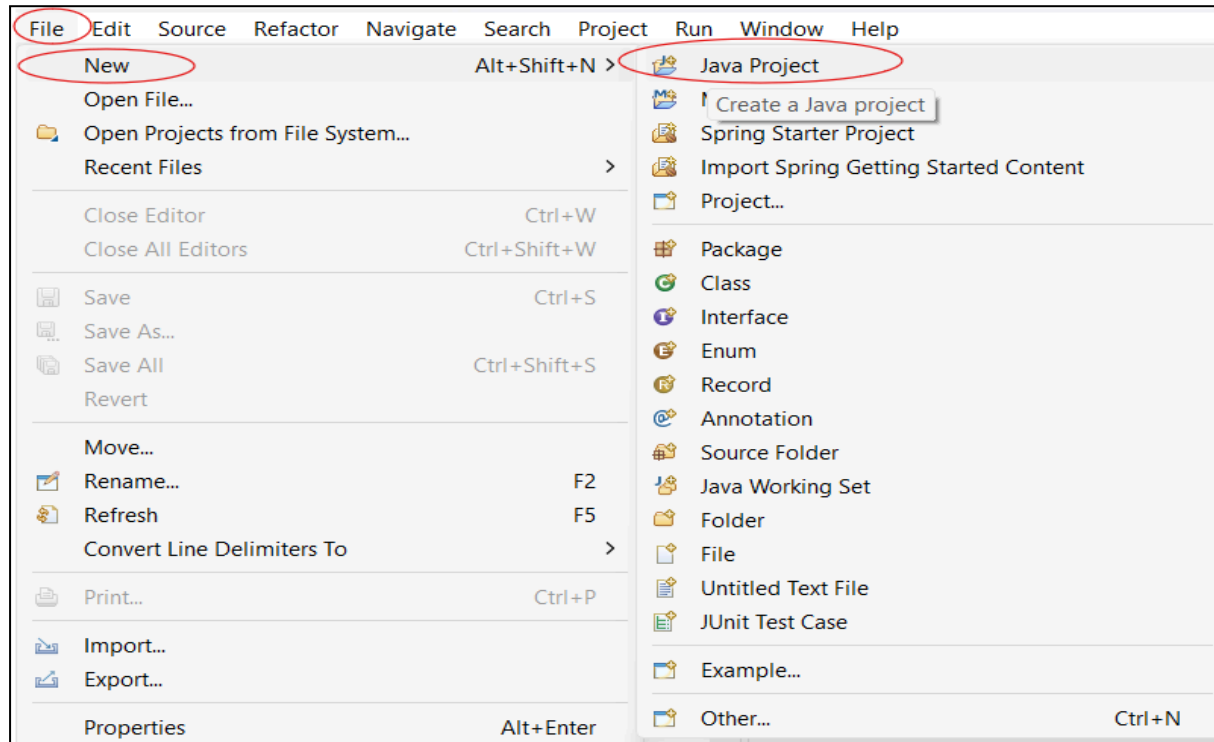


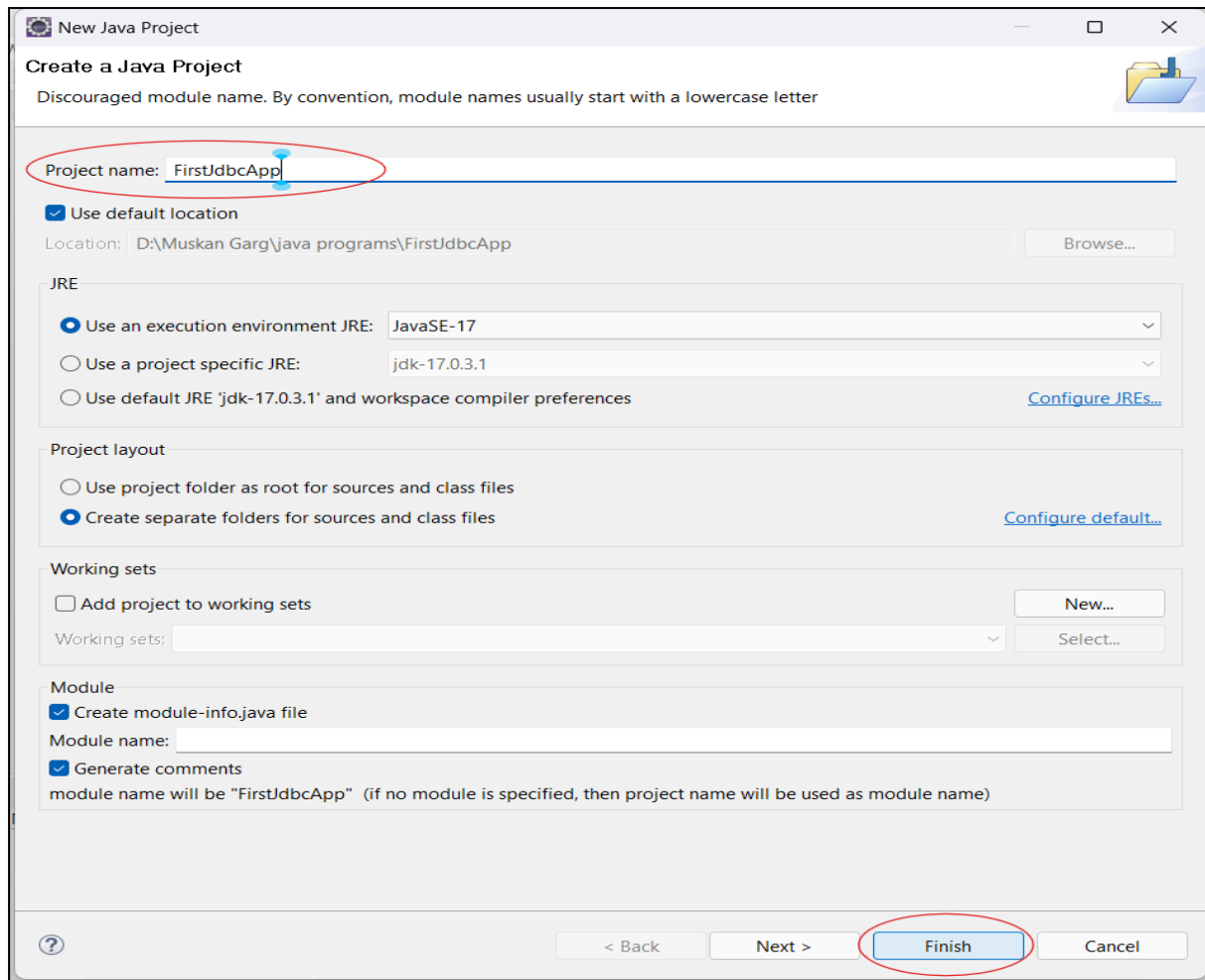
05: Setting Up JDBC Development Environment

The step-by-step guide to help you set up JDBC in your Java project:

Note: While we're using Eclipse IDE in this guide, feel free to use any IDE of your choice.

1. Create a Java Project: Start by creating a new Java project in your preferred IDE.





2. Download the JDBC Driver for MySQL:

- Visit the official MySQL connectors website to download the driver (JDBC Driver for MySQL (Connector)).

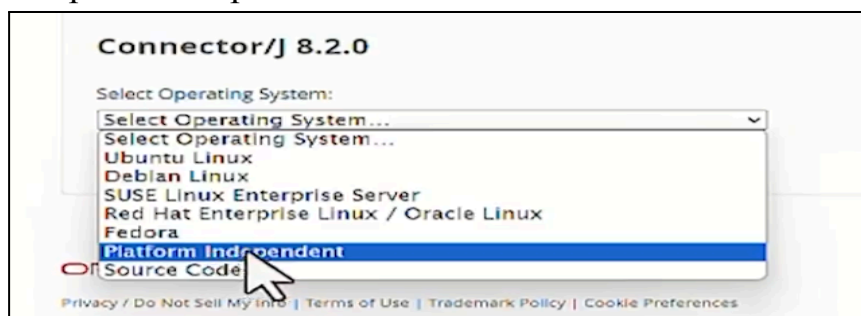
Link: <https://www.mysql.com/products/connector/>

The screenshot shows the MySQL website's "MySQL Connectors" page. The URL in the browser is `mysql.com/products/connector/`. The page features the MySQL logo and navigation links: `MYSQL.COM`, `DOWNLOADS`, `DOCUMENTATION`, and `DEVELOPER ZONE`. A secondary navigation bar includes `Products`, `Services`, `Partners`, `Customers`, `Why MySQL?`, `News & Events`, and `How to Buy`. On the left, a sidebar lists various products, with `MySQL Enterprise Edition` expanded to show options like `Datasheet (PDF)`, `Technical Specification`, `MySQL Database`, `Enterprise Backup`, `Enterprise HA`, `Enterprise Scalability`, `Enterprise Stored Programs`, `Enterprise Authentication`, `Enterprise TDE`, `Enterprise Encryption`, and `Enterprise Masking`. The main content area is titled `MySQL Connectors` and explains that MySQL provides standards-based drivers for JDBC, ODBC, and .Net. Below this, a section titled `Developed by MySQL` contains a table of connectors:

Developed by MySQL	
ADO.NET Driver for MySQL (Connector/NET)	Download
ODBC Driver for MySQL (Connector/ODBC)	Download
JDBC Driver for MySQL (Connector/J)	Download
Node.js Driver for MySQL (Connector/Node.js)	Download
Python Driver for MySQL (Connector/Python)	Download
C++ Driver for MySQL (Connector/C++)	Download

The `JDBC Driver for MySQL (Connector/J)` row is circled in red.

- Go to the "Select Operating System" section and choose the "Platform Independent" option.



- Download the "Platform Independent (Architecture Independent).ZIP Archive".

This screenshot shows the `Connector/J 8.2.0` download page with the `Select Operating System:` dropdown set to `Platform Independent`. Below the dropdown, there are two download options:

Download Option	Version	Size	Action
Platform Independent (Architecture Independent), Compressed TAR Archive (mysql-connector-j-8.2.0.tar.gz)	8.2.0	4.0M	Download
Platform Independent (Architecture Independent), ZIP Archive (mysql-connector-j-8.2.0.zip)	8.2.0	4.8M	Download

The `Download button for the ZIP Archive is highlighted with a mouse cursor. MD5 hashes and signature links are provided for each download option.`

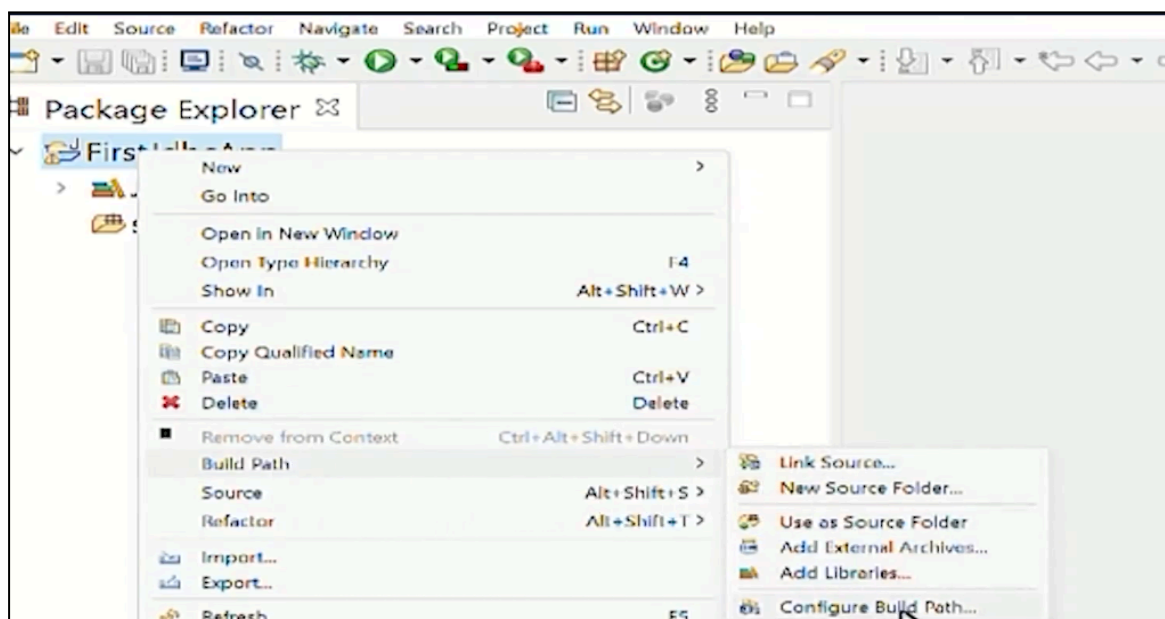
3. Extract the downloaded ZIP archive:

- Once the download is complete, extract the contents of the ZIP archive.
- After extraction, you will find the executable JAR file (e.g., mysql-connector-java-8.x.x.jar).

4. Add the JAR File to Your Java Project:

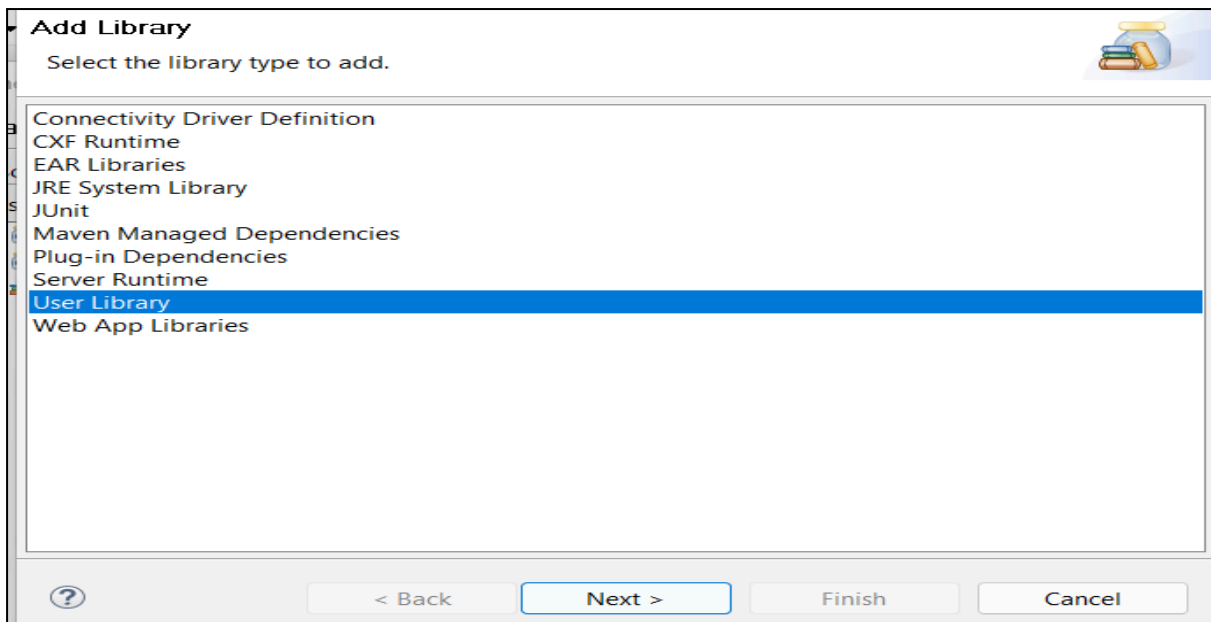
Open the Java project in which you want to add the JDBC driver.

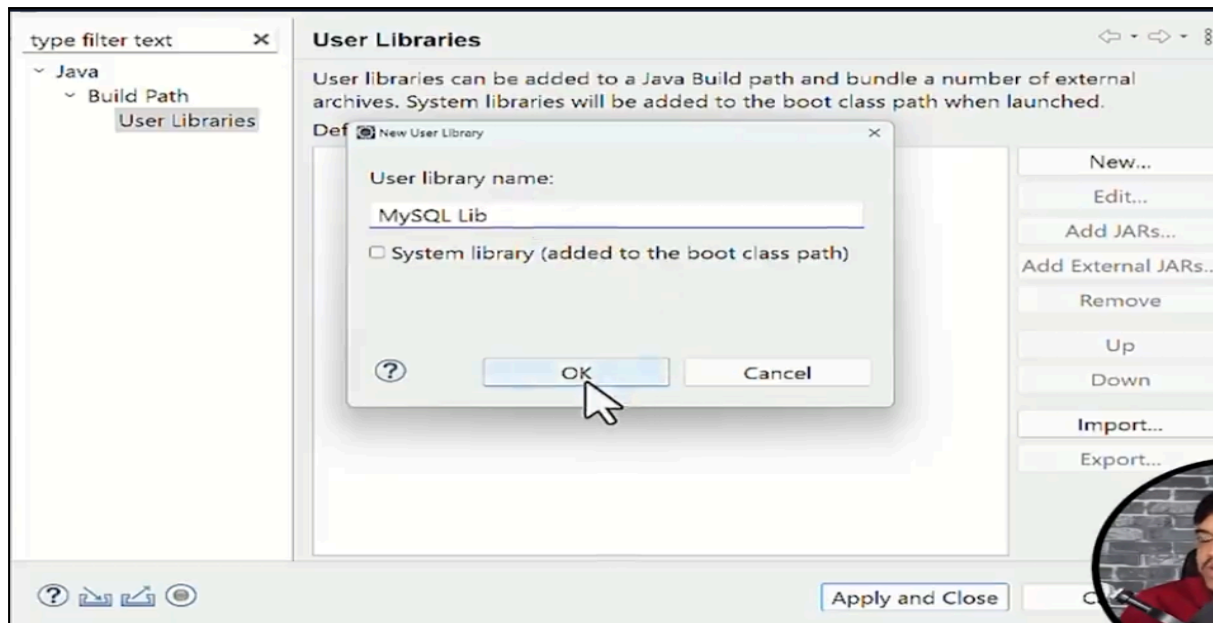
- **Configure Build Path:**
 - o Right-click on your project in the IDE.
 - o Go to **Build Path -> Configure Build Path**.
 - o A new window will open where you can configure the build path settings.



5. Create a User Library for the JDBC Driver (Optional):

- **New Library:**
 - o In the new window, you can create a new user library to keep the JAR file, making it reusable across different projects.
 - o Click on **Add Library -> User Library -> Next -> User Libraries -> New**.
 - o Give a name to the user library (e.g., MySQL Lib).

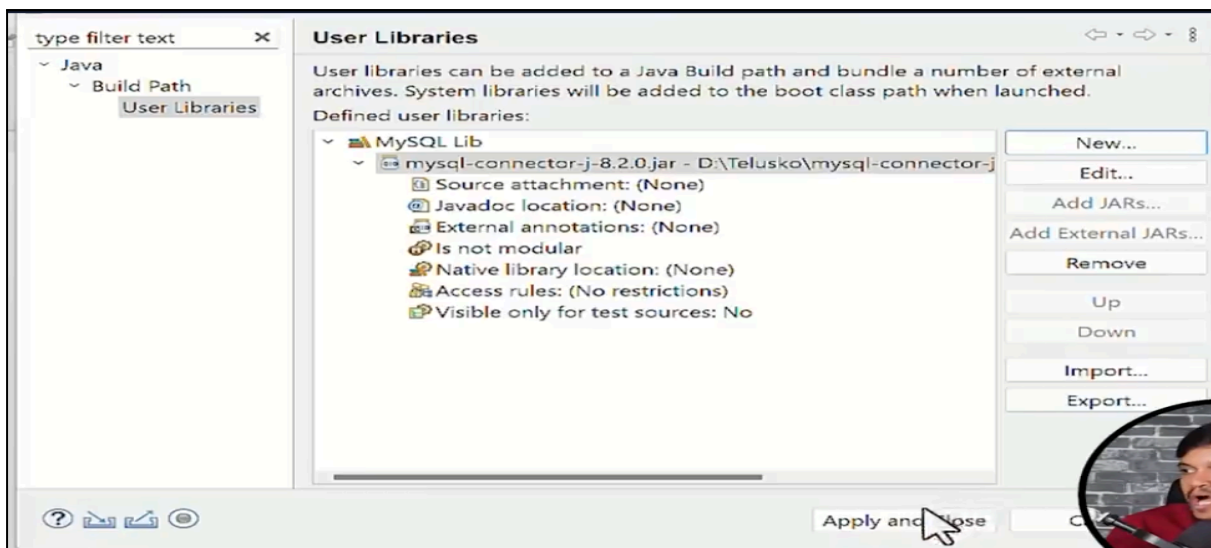
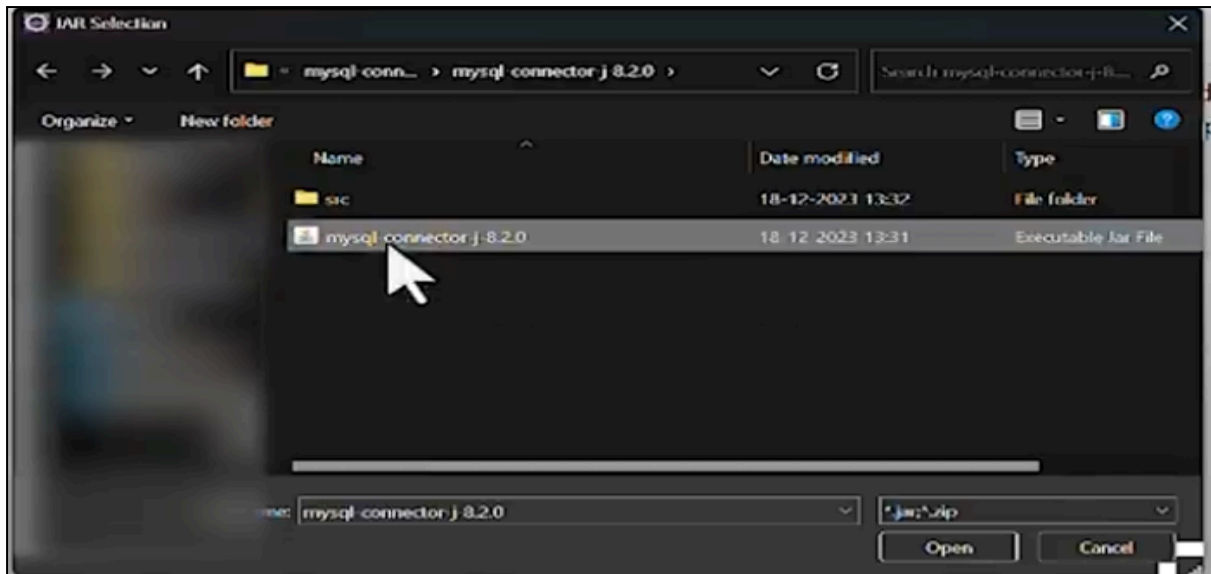




- **Add External JARs:**

- Select the newly created library.
- Click on **Add External JARs**.
- Navigate to the folder where you extracted the JAR file, select the mysql-connector-java-8.x.x.jar file, and click **Open**.
- Click **Apply and Close** to finish adding the JAR to your project.





6. Directly Add the JAR File (Alternative to User Library):

- **Add JAR:**
 - Instead of creating a user library, you can directly add the JAR file to your project.
 - Go to **Build Path -> Configure Build Path -> Libraries -> Add External JARs**.
 - Select the JAR file and apply the changes.

7. Create a New Java Class:

After setting up the JAR file, create a new Java class where you will write your JDBC code.

8. Import the JDBC Package:

In your Java class, import the necessary JDBC package to use JDBC classes and interfaces.